

Network Analysis Report

1. Introduction

This report analyzes a simple social network representing collaboration within a research group. Each member is modeled as a *node*, and each collaborative relationship is modeled as an *edge*. The goal is to construct the graph, compute key metrics—such as number of nodes, edges, degree distribution, and isolated members—and visualize the network using Python’s NetworkX library.

2. Network Construction

The network consists of five members: Stephen, Ama, Kojo, Yaw, and Dora. Relationships were defined as follows:

- Stephen → Ama, Kojo
- Ama → Kojo, Yaw
- Kojo → Yaw
- Dora → (no connections)

This creates a connected cluster of four individuals and one isolated member (Dora). NetworkX was used to build the graph, and a spring layout was applied to visualize the structure, as it positions highly connected nodes closer together.

3. Graph Metrics

3.1 Nodes and Edges

- Nodes: 5
- Edges: 4

3.2 Degree Distribution

- Stephen: 2
- Ama: 2
- Kojo: 2
- Yaw: 1

- **Dora: 0**

The distribution shows that Stephen, Ama, and Kojo form the core of the network. Yaw is less connected, while Dora is completely isolated.

3.3 Isolated Nodes

- **Isolated Member: Dora**

Identifying isolated nodes is important in group analysis, as they often represent individuals who may be disengaged or not integrated into team activities.

4. Visualization and Interpretation

The visualization reflects the structure clearly: a tight central cluster involving Stephen, Ama, and Kojo, Yaw connected at the edge, and Dora entirely separated. This helps illustrate collaboration patterns, highlight active members, and pinpoint individuals who may require more engagement.

5. Conclusion

The analysis shows a strongly connected core group and a single isolated member. The degree distribution and visualization provide insights into participation and collaboration patterns within the team. Python tools like NetworkX and Matplotlib make it easy to model, analyze, and interpret such structures, offering valuable understanding of how groups function.
